American Computer Science League

2021-2022 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A\,(B\ +\ \overline{C})}\ +\ \overline{A}\,\overline{B}(A\ +\ C)$$

A. $A + \overline{B}C$
B. $\overline{A} + \overline{B}\,\overline{C}$
C. $\overline{A} + BC$
D. $\overline{A} + \overline{B}C$
E. $A + BC$

2. Boolean Algebra

How many ordered triple(s) make the following Boolean expression TRUE?

$$(A\ \oplus\ (B\ +\ A)\,\overline{C}\,)\ +\ \overline{A}\,\overline{B}\,C$$

A. 0
B. 2
C. 3
D. 4
E. 6

3. Data Structures

Find the internal path length in the binary search tree for:

BEIJINGWINTEROLYMPICS

A. 91
B. 94
C. 100
D. 109
E. 110

4. Data Structures

Given an initially empty stack and the following commands on the stack, what will be the next item popped?

PUSH("C"), PUSH("A"), X=POP( ), PUSH("S"),
PUSH("S"), PUSH("I"), X=POP( ), X=POP( ),
PUSH("O"), PUSH("P"), X=POP( ), PUSH("E"),
PUSH("I"), X=POP( ), PUSH("A"), X=POP( ), X=POP( )

A. A
B. C
C. I
D. O
E. E

5. FSAs and Regular Expressions

Which of the following strings match the regular expression pattern:

[ACSL]*[^aeiou]*21.2

a. Ab212
b. ACSL21
c. ACSL442122
d. cmptr212
e. A202122
f. pgrmmng21

A. a, c
B. b, c
C. c, e
D. d, e
E. e, f

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1. 布尔代数

化简**以下布尔表达式：**

$$\overline{A\,(B\ +\ \overline{C})}\ +\ \overline{A}\,\overline{B}(A\ +\ C)$$

A. $A + \overline{B}C$
B. $\overline{A} + \overline{B}\,\overline{C}$
C. $\overline{A} + BC$
D. $\overline{A} + \overline{B}C$
E. $A + BC$

2. 布尔代数

有多少个有序三元组可以使以下布尔表达式结果为真？

$$(A\ \oplus\ (B\ +\ A)\,\overline{C}\,)\ +\ \overline{A}\,\overline{B}\,C$$

A. 0
B. 2
C. 3
D. 4
E. 6

3. 数据结构

请计算以下二叉搜索树的内部路径长度：

BEIJINGWINTEROLYMPICS

A. 91
B. 94
C. 100
D. 109
E. 110

4. 数据结构

给出一个初始为空的栈，并在此栈上执行以下命令。请问下一个将要出栈的元素是什么?

PUSH(“C”), PUSH(“A”), X=POP( ), PUSH(“S”),
PUSH(“S”), PUSH(“I”), X=POP( ), X=POP( ),
PUSH(“O”), PUSH(“P”), X=POP( ), PUSH(“E”),
PUSH(“I”), X=POP( ), PUSH(“A”), X=POP( ), X=POP( )

A. A
B. C
C. I
D. O
E. E

5. FSAs 和正则表达式

以下哪些字符串与**下方的**正则表达式相匹配：

[ACSL]*[^aeiou]*21.2

a. Ab212
b. ACSL21
c. ACSL442122
d. cmptr212
e. A202122
f. pgrmmng21

A. a, c
B. b, c
C. c, e
D. d, e
E. e, f